

Serial Capture and Pack Units SOP

One episode packs one unit. It starts with a plain unit at the unit spot, an open carton with its lid folded back, a rack of separators, a scanner, a log panel and an empty print slot. It ends with that unit standing in the carton behind its own separator, its serial read green and logged, and every tool back on its spot. Six episodes fill one carton. The sixth episode goes on to seal the carton and to put the serial-range label on it. One carton is six units, and one session is one carton.

The order never changes: scan the serial, log it, nest the unit with its separator, and on the sixth unit seal the carton and label it. No unit goes into the carton until its serial has read green and been logged. The carton is never sealed while a bay is still open, and the serial-range label is never taken until the carton is sealed.

The gripper does all the work. It takes each unit, holds the serial to the scanner window, presses the log button, sets the unit in the carton, stands a separator beside it, closes and tapes the lid, and puts the serial-range label on the carton end. The carton stays on its spot and is never lifted or turned during an episode.

Only one thing is held at a time. Nothing is passed from one grip to another, no grip is changed while an object is off the table, and nothing is dragged or slid along the table. Every pick lifts straight up off the surface before it moves anywhere.

All six units in a carton are the same product, and their serials are logged in the order they are scanned. The counts are fixed: one unit and one separator per episode, up to three scan tries, one log press, one press of 2 seconds on the tape and on the label, and six units to a carton. There is no second log press, no second tape strip, and no relabelling once the label is down. A unit, a separator, a tape strip or a label that lands on the table is left exactly where it falls, tagged, and the episode goes on. Nothing is ever tidied up on camera.

Setup

Finish both checklists before starting an episode.

Cell configuration

Environment camera: 90 mm.

cell_type: single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the unit spot, the separator rack, the carton, the scanner, the log panel, the print slot, the tape plate, the waste spot and the reject spot.

3. The serial and the scanner window both stay in frame for the whole scan, the log screen stays in frame and readable, and the inside of the carton stays in frame while a unit or a separator goes in.
4. The gripper is at home and open.
5. The tabletop is dry and holds nothing but the items in the Materials checklist.
6. The scanner is on. It gives one beep and a green light on a test read of a spare serial.
7. The log panel is on. Its screen is clear, its count reads zero, and its log button clicks back up on a test press.
8. The label printer is on and loaded. It gives one test label at the print slot, and that test label is taken away before the session starts.
9. The gripper reaches the unit spot, the scanner window, the log button, all six carton bays, the separator rack, the tape plate, the print slot and both carton end panels without stretching to a joint limit or folding in on itself. Bring a spot forward until this is true.
10. The paths from the unit spot to the scanner, from the scanner to the log panel, from the log panel to the carton, from the separator rack to the carton, and from the tape plate and the print slot to the carton are clear. Carry one unit and one separator along each path by hand during setup: they must clear the carton walls and the folded back lid. Widen the gaps until they do.

Materials checklist

1. Six plain units stand at the unit spot at the back left. Each one has one serial printed on one face, and each serial is different. They are the same size and shape, so one grip works on all six.
2. Each unit is square enough for the gripper to close on it squarely and hold it level. The serial is flat, dry and unbroken, so the scanner can read it in one look.
3. The serial faces up on every unit at the unit spot, so the gripper never has to turn a unit to find it.
4. One open carton stands at the carton spot at the back right. It holds six units in one row, its lid is folded right back and held there, and its mouth is open and clear.
5. The carton grips the table. Push it by hand during setup: if it slides or lifts, weigh it down or clamp it. Nothing pins the carton during the episode.
6. The carton is deep enough to hold a unit upright and shallow enough that a unit and a separator go straight down into it without being pushed.
7. Six flat separators stand on edge in the separator rack at the back middle, each one tall enough to stand above the units and thin enough to slide down beside one. The top edge of each separator stands clear of the rack, so the gripper has an edge to close on.
8. One tape plate lies flat at the front left with pre cut tape strips on it, well more than one strip, and the corner of the next strip lifted off the plate. A strip lying flat with no lifted corner is not usable: the gripper cannot lift the corner and peel the strip at the same time.

9. The print slot at the middle right is empty at the start of the session. The serial-range label comes out here and nowhere else.
10. The waste spot at the front middle is empty or only part full. It takes bad tape strips, bad labels and backing paper and nothing else.
11. The reject spot at the front right is empty. It takes units that will not read and nothing else.
12. A clear gap of at least a gripper's width sits around the carton on every side, so a unit and a separator go straight down into it and come away without touching a carton wall or the folded back lid.

Workspace layout

Three rows: the units, the separators and the carton at the back, the scanner, the log panel and the print slot in the middle, the tape plate and the two drop spots at the front. Each unit travels forward to the scanner, across to the log panel, then back to the carton.

Back row, left to right:

1. **Unit spot**, back left. The plain units stand here, serial face up. It empties as the carton fills.
2. **Separator rack**, back middle. The separators stand on edge here, one per unit.
3. **Carton spot**, back right. The open carton stands here with its lid folded back. It fills as the unit spot empties.

Middle row, left to right:

1. **Scan spot**, middle left, the scanner standing with its window facing the front of the table.
2. **Log panel**, middle centre, the screen above and the log button below it. Everything that happens to a serial happens here.
3. **Print slot**, middle right, beside the carton. The serial-range label comes out flat here after the sixth log press.

Front row, left to right:

1. **Tape plate**, front left. It holds the tape strips flat with the corner of the next strip lifted.
2. **Waste spot**, front middle, the strip of bare table that takes bad tape strips, bad labels and backing paper.
3. **Reject spot**, front right. It takes a unit that will not read.

No carry passes over the open carton on the way to somewhere else, no carry passes over the log panel or the scanner, and nothing travels to the far side of the table.

Vocabulary

Unit: one item that has to be scanned, logged and packed.

Serial: the code printed on one face of the unit. Every unit has one and no two are the same.

Serial face: the face of the unit that carries the serial. It faces up at the unit spot.

Scanner: the fixed reader at the scan spot. Its window is its reading face.

Scan gap: the 5 to 10 cm of daylight between the serial and the scanner window, held for the whole scan try.

Scan try: one hold of the serial in front of the window, square to it, for up to 3 seconds. Three tries is the limit.

Good read: one beep and a green light from the scanner. Nothing else counts as a read.

Log panel: the screen and the button at the middle of the table. The screen shows the serial just read, and the button writes it into the log.

Log button: the button below the screen. One press per unit, and the button comes back up after the press.

Log line: the new line on the screen holding the serial just logged.

Count: the number on the screen. It goes up by one on every good log press, from zero to six.

Carton: the open box at the carton spot. It holds six units in one row and it stays where it is.

Bay: one unit place in the carton. The next open bay is the first empty one, filling from the left end of the carton to the right end, ending at the right end as bay six.

Separator: one flat card that stands between two units so they never touch. One goes in behind each unit.

Nested unit: a unit standing upright in its bay, serial face up, with its separator standing beside it and nothing leaning.

Lid: the flap folded back behind the carton. It turns down over the mouth to close it.

Seam: the line where the closed lid meets the front wall of the carton. The tape goes across the middle of it.

Tape strip: one pre cut strip of tape from the tape plate, sticky side down.

Lifted corner: the corner of the next tape strip or of the label raised off its backing. It is the grasp: the gripper cannot lift a strip or a label that lies flat.

Sealed carton: the lid turned down flat and one tape strip pressed across the middle of the seam, with no lifted corner and no gap into the carton.

Serial-range label: the one label printed after the sixth log press. It shows the first serial and the last serial in the carton.

First serial: the serial of the unit logged first in this carton. **Last serial:** the serial of the unit logged sixth.

Level label: a label whose long edges run parallel to the long edges of the carton end panel, within 5 degrees.

Flat label: a label pressed down with no lifted corner, no crease and no bubble under it.

Press: the closed gripper pushed onto the tape or the label for 2 seconds, from the middle out to each edge.

Full carton: six units, one in every bay, each with its separator, and the count on the screen reading six.

Hold: a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.

Bad tape strip: a strip that is torn, folded onto itself, doubled with a second strip, or stuck down somewhere it should not be. It goes to the waste spot.

Reject unit: a unit that will not read after three scan tries. It goes to the reject spot.

Stable placement: the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.

Spill: a unit, a separator, a tape strip or a label that ends up outside the carton, the waste spot or the reject spot. It is left where it falls and cleared in the reset, never on camera.

Home: the resting pose, clear of the table with the gripper open.

Steps

Step 1: Scan the serial

Goal: the scanner gives one good read of the serial on one unit.

The carton stays open and untouched for this whole step. Nothing goes into the carton before the read.

1.1 Take one unit

Close on one unit at the unit spot, square to it, so it hangs level with the serial face up.

Lift it straight up off the table before any sideways move.

Take one unit per episode. Never lift two at once, and never drag or slide a unit across the table.

Check: one unit hangs level in the gripper and nothing else at the unit spot has moved. If two units came away together, set both back on the unit spot and take one again. Make no more than two extra tries.

1.2 Show the serial to the scanner

Carry the unit to the scanner and hold the serial in front of the window at the scan gap, square to the window.

Hold it still for up to 3 seconds.

If the scanner beeps once and shows green, go to Step 2.

If nothing reads within 3 seconds, move the unit 2 to 3 cm closer to the window or farther from it and try once more. **Three scan tries in total, and no more.**

Only the serial face is shown to the scanner. Never turn the unit to show another face, and never scan a unit twice.

Check: the scanner has given one beep and a green light for this unit. If there is no good read after three tries, or the scanner shows red, put the unit at the reject spot, end the episode, and start a new one. The unit does not go in the carton.

Step 2: Log the serial

Goal: the serial just read stands on the screen as a new log line and the count has gone up by one.

The unit stays in the gripper for this whole step. It is not set down and it is not put in the carton yet.

Carry the unit to the log panel and look at the screen. The serial just read shows there.

With the unit still held, bring the gripper over the log button and press it once, straight down, until it clicks.

Lift straight up off the button and let it come back up.

One press per unit. Never press the button twice for the same unit, and never press it with no unit in the gripper.

Check: the screen shows the serial just read on a new log line and the count has gone up by exactly one. If the screen shows the wrong serial, put the unit at the reject spot, end the episode, and start a new one. If the count did not move, press the button once more, and once only. If it still does not move, end the episode and check the log panel during reset.

Step 3: Nest the unit with its separator

Goal: the unit stands upright in the next open bay, serial face up, with one separator standing beside it, and the carton still on its spot.

3.1 Set the unit in its bay

Carry the unit from the log panel to the carton, level the whole way, and bring it over the next open bay.

Fill the bays in order, from the left end of the carton to the right end, ending at the right end as bay six. Never skip a bay and never stand a unit on one already in the carton.

Lower the unit into the bay until it is standing, serial face up, then open the gripper and lift clear. Open the gripper low, no more than 2 cm above the carton mouth, and never drop the unit in.

Check: the unit stands in the right bay, serial face up, and does not rock or lean. If it is off place or leaning, lift it clear and place it again. Make no more than one correction.

3.2 Stand a separator beside it

Close on the top edge of the next separator in the rack and lift it straight up out of the rack.

Carry it to the carton and lower it straight down beside the unit just nested, on the side the next unit will come from, until it stands on the carton floor.

Open the gripper and lift clear. **One separator per unit.** Never put two in at once and never push a separator down with the unit already leaning on it.

The separator goes in after the unit, never before it. A separator dropped in first leaves no room to lower the unit straight down.

Check: the separator stands upright on the carton floor, its full face between the unit just nested and the next open bay, and the unit still stands upright. If the separator leans, sits on top of a unit, or has pushed a unit out of place, lift it clear and place it again. Make no more than one correction.

Step 4: Seal the carton

Goal: the lid lies flat over the carton mouth and one tape strip is pressed across the middle of the seam.

Count the nested units and read the count on the screen. If the carton holds fewer than six units, go to Step 6.

If this was the sixth unit, go on to 4.1.

The carton is sealed only on the sixth unit, and only once every bay holds a unit and a separator.

4.1 Close the lid

Close on the middle of the free edge of the folded back lid.

Turn the lid down over the carton mouth in one move, keeping it level, until it lies flat on the carton and its free edge meets the front wall.

Open the gripper, close it empty, and press the lid down along the seam so it stays flat.

Check: the lid lies flat over the whole mouth, no unit and no separator stands above the mouth holding it up, and the seam runs straight across the front wall. If the lid rides up on a unit or a separator, lift the lid back, seat that unit or separator again, and close the lid once more. Make no more than one correction.

4.2 Tape the seam

Close on the lifted corner of the next tape strip on the tape plate.

Peel the strip off in one move, up and toward the front of the table, keeping the sticky side down the whole way.

Carry it to the carton and hold it 1 to 2 cm above the middle of the seam, still clear of it.

Turn it until it runs square across the seam, then lower it straight down until the whole sticky side is touching, and open the gripper.

Close the gripper and press the strip down for 2 seconds, starting at the middle and working out to each end.

Once the strip is down it is not slid, nudged or turned. A strip that landed crooked is taken off, not pushed straight.

Check: one tape strip lies flat across the middle of the seam, holding the lid to the front wall, with no lifted corner and no gap into the carton. If a corner lifts, press it down once more. If the strip is torn, folded onto itself, or has come away, lay it on the waste spot and peel one more strip. Make no more than one extra strip.

Step 5: Put the serial-range label on the carton

Goal: the serial-range label lies flat and level on the carton end panel and reads the first serial and the last serial of this carton.

The label is printed by the sixth log press and waits flat in the print slot. It is not taken until the carton is sealed.

5.1 Take the label from the print slot

Close on the lifted corner of the label in the print slot.

Peel it off its backing in one move, up and toward the front of the table, keeping the printed side up the whole way.

Take one label per carton. Never print a second label and never take a label from anywhere but the print slot.

Check: one whole label hangs from the gripper, printed side up, and the print slot is empty. If the label tears or folds onto itself, lay it on the waste spot, end the episode, and print a fresh label during reset.

5.2 Place it and press it flat

Carry the label to the carton end panel and hold it 1 to 2 cm above the middle of the panel, still clear of it.

Turn the label until its long edges run parallel to the long edges of the end panel, within 5 degrees.

Make no more than two small moves to line it up. **The label is lined up before it touches, not after.**

Lower it straight down until the whole sticky side is touching, then open the gripper and lift clear.

Close the gripper and press the label down for 2 seconds, starting at the middle and working out to each of its four edges.

Check: the label lies flat and level on the end panel, no corner lifts, no crease shows, and no part of it hangs over an edge of the panel. If one corner lifts or a crease shows, press that

corner or crease down once more. Make no more than one correction. A label that is crooked or overhanging is left as it is and tagged: it is not peeled off and it is not covered with a second label.

5.3 Close out the full carton

Read the label. It shows the first serial and the last serial of this carton.

Close on the middle of the front wall of the carton, push the carton straight back until its far wall touches the back edge of the table, and open the gripper.

Check: a sealed carton of six sits pushed back against the table edge, taped across the seam, with the serial-range label flat on its end panel. A carton holding fewer than six units is left exactly where it stands and is never sealed, labelled or pushed back.

Step 6: End the episode

Check the unit: read green, logged, standing in its bay with the serial face up and its separator beside it.

Check the log panel: the screen shows the serial on a new line and the count has gone up by exactly one.

Check the carton: on its spot, every filled bay holding one unit and one separator, and no unit leaning.

Check the tools: the tape plate flat with the corner of the next strip lifted, and the print slot empty unless this was the sixth unit.

On the sixth unit, also check the seal and the label: the lid flat, one tape strip across the middle of the seam, the label flat and level on the end panel, and the carton pushed back.

Fix a failed check only with the fix written in its step. If no fix is written, or the fix would need an earlier step run again, end the episode and reset instead.

Go home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Leave a sealed carton where it stands, take it away for the next stage, and stand a fresh open carton at the carton spot with all six bays empty and its lid folded right back.
2. Clear the log panel back to zero, so the count starts at zero for the new carton and the screen holds no old log line.
3. Refill the separator rack so it holds six separators standing on edge, each top edge clear of the rack. Replace a separator that is bent, torn or damp.
4. Lift the corner of the next tape strip on the tape plate off the plate, so the gripper has an edge to close on. Replace a plate holding fewer than two strips.

5. Empty the print slot. Throw away any label left in it, and print a fresh label only from a fresh count of six.
6. Empty the waste spot into the bin, so only the tape and the labels from the next episode can land there.
7. Empty the reject spot. Set any reject unit aside for a person to look at, and do not put it back on the unit spot.
8. Top up the unit spot so it holds enough plain units to finish the carton, each standing serial face up, each serial different from every other one in the carton.
9. Wipe any loose tape, backing paper or dust off the tabletop, out of the carton and out of the separator rack, and dry them.
10. Check that the carton still grips the table, no unit is crushed or wet on its serial face, the scanner still reads a spare serial, the log button still clicks back up, and the printer still gives a test label.
11. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

Visible cue: a unit goes into the carton before it has been read green, or before the log button has been pressed for it; the separator goes in before the unit; the lid is closed before the sixth unit is nested; the serial-range label is taken before the carton is taped; or a step is skipped.

SOP rule broken: Steps 1 to 5, scan, log, nest with a separator, then seal and label on the sixth unit, in that order.

Coaching note: scan it, log it, nest it, then the separator. Seal and label only on the sixth unit.

Violation: Scan skipped

Visible cue: a unit goes from the unit spot toward the log panel or the carton with no scan try at the scanner window.

SOP rule broken: Step 1.2, every unit is read before it is logged and packed.

Coaching note: every unit meets the window. No unit is logged or nested unread.

Violation: Wrong face shown, or a unit scanned twice

Visible cue: a face with no serial on it, or a unit already logged in this carton, is held in front of the scanner window.

SOP rule broken: Step 1.2, the serial face is shown to the window at the scan gap, square to it, once per unit.

Coaching note: show the serial face, and show each unit once. A repeat scan puts the same serial in the log twice.

Violation: More than three scan tries

Visible cue: the serial is held in front of the window a fourth time instead of stopping at three.

SOP rule broken: Step 1.2, three scan tries in total, and no more.

Coaching note: the count is fixed at three. After the third, the unit goes to the reject spot and the episode ends.

Violation: Log press wrong or missing

Visible cue: the log button is not pressed for a unit that was read green; it is pressed twice for the same unit; it is pressed with no unit in the gripper; or the unit is set down before the press.

SOP rule broken: Step 2, one press per unit, made with the unit still held, straight down until it clicks.

Coaching note: hold the unit, press once, let the button come back up. One unit, one press.

Violation: Logged with no good read

Visible cue: the log button is pressed after a red light, or with no beep and no green light for that unit.

SOP rule broken: Steps 1.2 and 2, a serial enters the log only after one beep and a green light.

Coaching note: wait for the beep and the green before the button. A logged serial that was never read breaks the whole carton.

Violation: Nested with no good read or no log

Visible cue: a unit is lowered into a bay after a red light, or before the count on the screen has gone up for it.

SOP rule broken: Steps 1.2, 2 and 3.1, a unit enters the carton only after a good read and a good log press.

Coaching note: green, then logged, then packed. Nothing enters the carton out of that order.

Violation: Wrong bay or fill order broken

Visible cue: the unit goes into a bay that is not the next open one, a bay is left open while a later bay is filled, or a unit is stood on one already in the carton.

SOP rule broken: Step 3.1, the next open bay, filling from the left end of the carton to the right end.

Coaching note: fill from the left end to the right end in order. Never skip a bay.

Violation: Unit nested serial face down

Visible cue: the unit stands in the bay with the serial against a wall, against a separator, or turned face down.

SOP rule broken: Step 3.1, the unit stands with the serial face up.

Coaching note: the serial must be easy to read from above once the unit is in its bay.

Violation: Unit dropped into the carton

Visible cue: the gripper opens more than 2 cm above the carton mouth, the unit falls the last part of the way, or it lands well off its bay.

SOP rule broken: Step 3.1, lower the unit into the bay and open the gripper low.

Coaching note: carry it down into the bay, then let go. Lower until it is standing, then open.

Violation: Separator missing, doubled, or put in first

Visible cue: a unit is nested with no separator beside it, two separators go in for one unit, or a separator is lowered into the bay before the unit.

SOP rule broken: Step 3.2, one separator per unit, placed after the unit.

Coaching note: unit first, then one separator beside it. The separator has nothing to lean on until the unit is standing.

Violation: Separator placed wrong

Visible cue: the separator leans, sits on top of a unit, stands outside the carton, or pushes a nested unit out of place.

SOP rule broken: Step 3.2, the separator stands upright on the carton floor with its full face between the unit and the next open bay.

Coaching note: lower it straight down beside the unit until it stands on the floor. A leaning separator lets the next unit touch this one.

Violation: Carton sealed early or not sealed

Visible cue: the lid is turned down while a bay is still open or the count on the screen reads fewer than six, or the sixth unit is nested and the episode ends with the carton still open.

SOP rule broken: Step 4, the carton is sealed on the sixth unit and only on the sixth unit.

Coaching note: count the bays and read the count on the screen. Six closes the carton, five does not.

Violation: Lid closed on a standing unit or separator

Visible cue: the lid rides up on a unit or a separator instead of lying flat, or it is pressed down onto one that stands above the mouth.

SOP rule broken: Step 4.1, the lid lies flat over the whole mouth with nothing holding it up.

Coaching note: if the lid will not lie flat, something inside is standing proud. Seat it again before closing.

Violation: Seam taped wrong

Visible cue: no tape strip goes on, more than one strip goes on, the strip misses the middle of the seam, runs along the seam instead of across it, is left with a lifted corner, or leaves a gap into the carton.

SOP rule broken: Step 4.2, one strip laid square across the middle of the seam and pressed flat.

Coaching note: one strip, square across the middle, pressed from the middle out. One strip seals it, two do not seal it better.

Violation: Bad tape strip used

Visible cue: a strip that is torn, folded onto its own sticky side, or doubled is carried to the carton and put down on the seam.

SOP rule broken: Step 4.2, a bad strip goes to the waste spot and one more strip is peeled.

Coaching note: a bad strip is never used. It goes on the waste spot and a fresh one is peeled.

Violation: Wrong label, or a label taken early

Visible cue: the label is taken from the print slot before the carton is taped, a second label is printed or taken for the same carton, or a label is taken from anywhere but the print slot.

SOP rule broken: Step 5.1, one label per carton, taken from the print slot after the seal.

Coaching note: seal first, then take the one printed label. One carton, one label.

Violation: Label not level, or badly placed

Visible cue: the label sits more than 5 degrees out of square with the long edges of the end panel after it is pressed, hangs over an edge of the panel, or is put on the lid, the front wall or a unit instead of the end panel.

SOP rule broken: Steps 5.1 and 5.2, the label goes flat and level on the carton end panel.

Coaching note: square it while it is still 1 to 2 cm above the panel, before it touches.

Violation: Label left lifting or creased

Visible cue: at the end of Step 5 a corner of the label stands off the panel, or a crease or bubble is left under it.

SOP rule broken: Step 5.2, the label lies flat with no lifted corner and no crease.

Coaching note: press from the middle out to each edge. One extra press on a lifted corner, and that is all.

Violation: Press skipped or short

Visible cue: the tape strip or the label is let go and the gripper moves away with no closed gripper press, or the press is shorter than 2 seconds.

SOP rule broken: Steps 4.2 and 5.2, press for 2 seconds from the middle out to each edge.

Coaching note: hold the full 2 seconds. A quick tap lets the tape or the label lift later.

Violation: Tape or label pushed into place

Visible cue: the tape strip or the label is slid, nudged or turned after it has touched the carton, or a second label is laid over a first one.

SOP rule broken: Steps 4.2 and 5.2, once it is down it is not moved and it is not covered.

Coaching note: nothing on this station is pushed into line. A crooked label is tagged and left, not fixed on camera.

Violation: Full carton not closed out

Visible cue: the carton is sealed and labelled and the episode ends with it still forward of the table edge, or the carton is pushed back while it holds fewer than six units.

SOP rule broken: Step 5.3, push the carton back on the sixth unit and only on the sixth unit.

Coaching note: count the bays. Six closes the carton out, five does not.

Violation: Bad tape, bad label or reject unit put in the wrong place

Visible cue: a bad tape strip or a bad label is left on the table, put in the carton, or put at the reject spot; a reject unit is put on the waste spot, back on the unit spot, or in a bay; or a good unit is put at the reject spot.

SOP rule broken: Steps 1.2, 4.2 and 5.1, bad tape and bad labels go to the waste spot and failed units go to the reject spot.

Coaching note: tape and labels to the waste spot, units to the reject spot. Neither goes back into the flow.

Violation: Spill cleaned up

Visible cue: after a unit, a separator, a tape strip or a label lands somewhere it should not, the gripper is used to sweep, push, wipe or pick it up.

SOP rule broken: Steps 1 to 5, a spill is left where it falls and is cleared in the reset.

Coaching note: cleanup moves are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

Visible cue: an object moves from one grip to another without being set down first, two objects are held at once, a grip is shifted or taken again while the object is off the table, a unit or the carton slides along the table with no daylight under it, a unit hangs tilted because it was taken off square, or an object is turned over in the air.

SOP rule broken: Steps 1 to 5, one object at a time, lifted straight up off the surface before any sideways move, gripped square, and never passed between grips.

Coaching note: set it down to change the grip, and lift before the carry. A unit that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit passed

Visible cue: a unit, a separator, the carton, the tape plate or the lid is shoved, nudged or slid into place instead of being lifted clear and placed again, or a failed check is tried again more often than its step allows.

SOP rule broken: Steps 1 to 5, a bad placement is fixed by lifting the object clear and placing it again, and each check allows only the fix it states.

Coaching note: nothing on this station is pushed into line. If the grip or the approach is forcing the same fix again and again, let review flag it and move on.

Violation: Carry over a working spot

Visible cue: a unit, a separator, a tape strip or a label is carried over the open carton on its way to somewhere else, over the log panel, or over the scanner window, or a carry crosses the far side of the table.

SOP rule broken: Workspace layout, nothing is carried over the open carton, the log panel or the scanner, and nothing travels to the far side of the table.

Coaching note: go around, not across. A dropped unit over an open carton spoils a whole six.

Violation: Dropped, collided, or knocked over

Visible cue: under command, the gripper or a carried object hits the carton, the folded back lid, the separator rack, the scanner, the log panel or a nested unit and knocks, tips, slides or

moves it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.

SOP rule broken: Steps 1 to 5, keep each object on a clear path and let go only after a settled placement.

Coaching note: watch the whole path, not just the target.

Violation: Work outside the camera frame

Visible cue: a scan try, a log press, a bay placement, a separator placement, a lid close, a tape press or a label press happens partly or wholly outside the environment camera frame, or the log screen is out of frame when the count changes.

SOP rule broken: Steps 1 to 5, every move happens inside the camera frame, and the screen is readable when it changes.

Coaching note: the work has drifted outside the framed area. Set the spots back to their setup places.

Violation: Wrong episode ending

Visible cue: the episode ends with a unit still in the gripper, a unit unscanned, unlogged or unnested, a separator missing, a full carton not sealed, labelled or pushed back, short of home, or the gripper not fully open.

SOP rule broken: Step 6, check the unit, the log panel, the carton and the tools, then go home with the gripper open and stop recording.

Coaching note: check first. Going home is the last thing that happens before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Hardware fault: gripper failure, drift, controller caused collision, or motor error.

A spot sits outside comfortable reach, so the move stalls at a joint limit or has to fold in on itself to work the spot. Move the spot and run the Setup checklists again.

Scanner fault: no beep and no light on a flat, clean, fully shown serial after three scan tries, and the same failure on a test read of a spare serial. Check the scanner before the next episode.

Log panel fault: the screen freezes, shows the wrong serial for a good read, or the count does not move on a full press that the button answers. Check the panel before the next episode.

Printer fault: no label at the print slot after the sixth good log press, or a label that comes out blank, smeared or unreadable.

Bad unit: one whose serial is crushed, torn, smeared or wet, so it cannot be read.

Bad carton: one that slides or lifts under a correct placement, whose lid will not fold back or lie flat, or that will not sit flat on the table.

Bad separator: one that is bent, torn or damp, or that will not stand upright on the carton floor.

Bad tape plate: one whose next corner will not stay lifted, or whose strips have lost their stick.

Unit spot holding too few units to finish the carton, separator rack holding too few separators, or a tape plate holding too few strips. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take one unit from the unit spot
2. Show the serial to the scanner
3. Press the log button
4. Set the unit in the next open bay
5. Take one separator from the rack
6. Stand the separator beside the unit
7. Close the carton lid
8. Peel one tape strip from the tape plate
9. Press the tape strip across the seam
10. Take the serial-range label from the print slot
11. Place the label on the carton end panel
12. Press the label flat
13. Push the sealed carton back to close it out
14. Go home and end the episode